Deep Learning-Enabled Digital Twins with Quantified Uncertainty for Risk-Averse Management of Type 1 Diabetes

Md Azharul Islam

Department of Mechanical and Aerospace Engineering
University at Buffalo

Ph.D Research Proposal Presentation

December 1, 2025

PCELab
Predictive Computational Engineering

Table of contents

1. Motivation
2. Bayesian Temporal Convolutional Network (B-TCN)
3. B-TCN Model Discovery Using Population Based Data
4. Future Plan with Digital Twin for Diabetes

Outline

1. Motivation
2. Bayesian Temporal Convolutional Network (B-TCN)
3. B-TCN Model Discovery Using Population Based Data
4. Future Plan with Digital Twin for Diabetes

Motivation: Digital Twin (DT)

- **Diabetes and prediabetes are extremely common**
  In the United States, about 38–38.4 million people (≈ 1 in 10) are living with diabetes, and nearly 100 million adults (≈ 1 in 3) have prediabetes (blood sugar higher than normal but not yet diabetes). lifelong insulin dependence for (≈ 1 1.6M) patients in U.S.

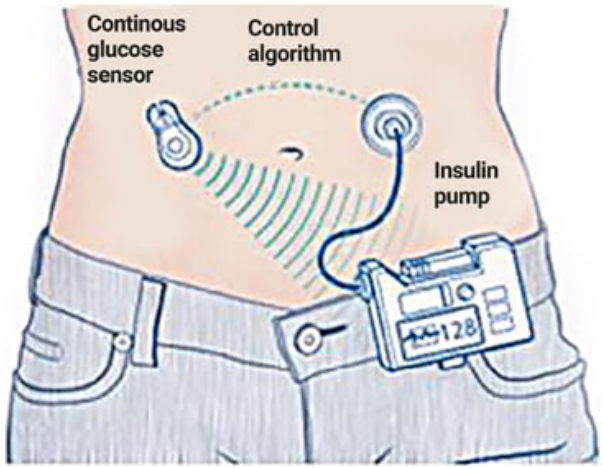
- **Motivation for Digital Twin (DT)**

To use artificial pancreas systems safely in everyday life, we need uncertainty-aware digital twins that can:

(i) Recognize when sensor data are noisy or unreliable,

(ii) Account for patient-to-patient variability, and

(iii) Support risk-averse, personalized insulin dosing.



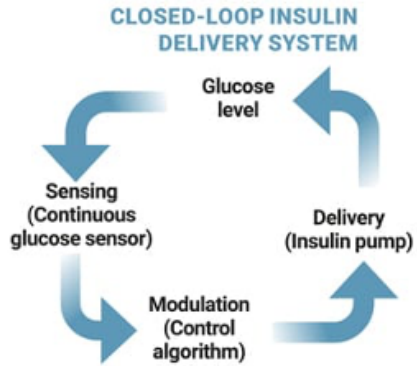


Illustration of a closed-loop artificial pancreas system.

Motivation: Enabling Artificial Pancreas

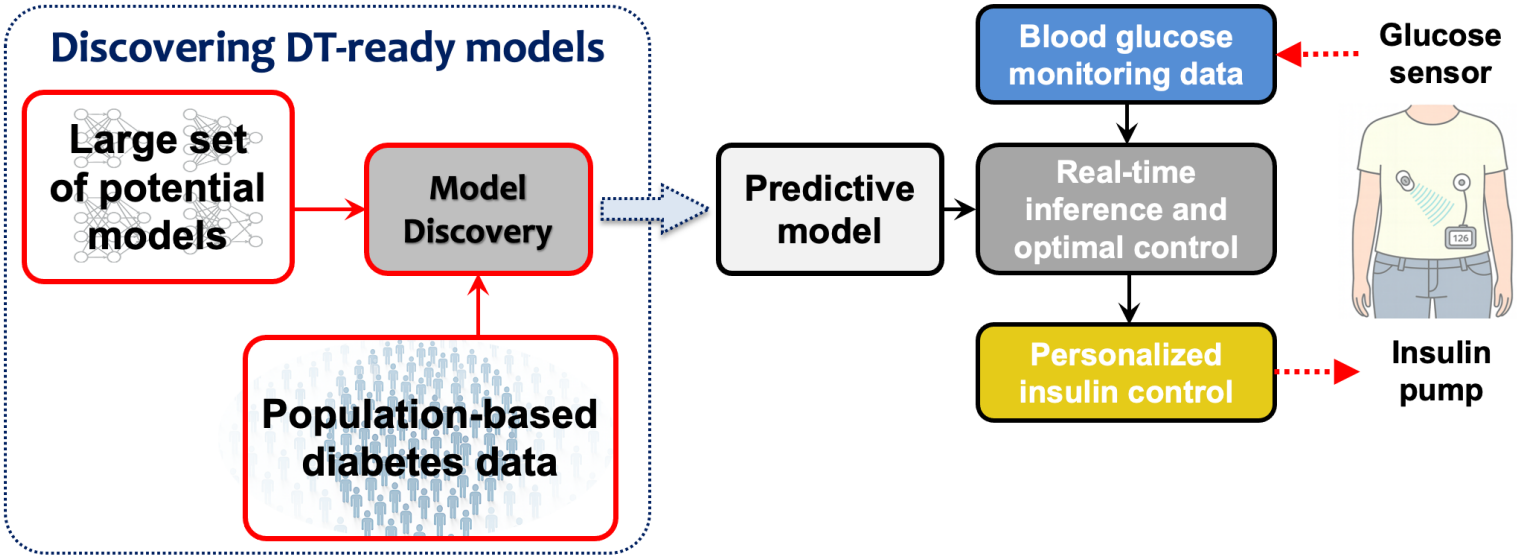


Illustration of a closed-loop artificial pancreas system.

Emphasis:

- **(i)** Validation and Uncertainty quantification in deep learning for time series.
- **(ii)** Patient-specific data accumulated noise from sensors and data.
- **(iii)** Risk-averse control: optimize insulin delivery while mitigating uncertainty.

Data source:

- **(i)** ODE-based blood glucose model.
- **(ii)** FDA-approved in silico patients: variable meal sizes, insulin dosing, age groups, and simulated glucose readings over two years.
- **(iii)** Clinical datasets: 300 T1D patients collected through the Tidepool Big Data Donation Project .

Outline

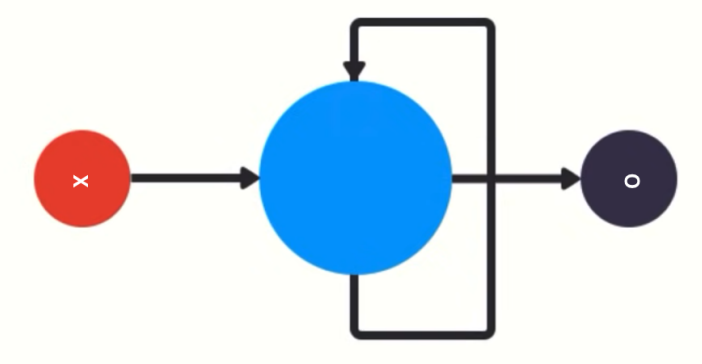
1 Motivation

2 Bayesian Temporal Convolutional Network (B-TCN)

3 B-TCN Model Discovery Using Population Based Data

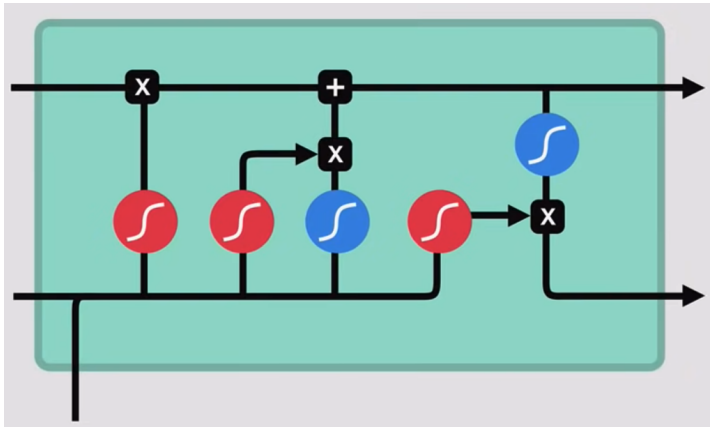
4 Future Plan with Digital Twin for Diabetes

Deep Learning for Time Series



(a)RNN

- Suffers from vanishing / exploding gradients.
- Hard to learn long-range dependencies.
- Lower accuracy than TCN and LSTM.



(b) LSTM

- Gating but adds many parameters and states.
- Still sequential in time: slow for long sequences.
- Can still forget very long-term context and uses large memory.

Solution: Temporal Convolutional Networks (TCN) [1]

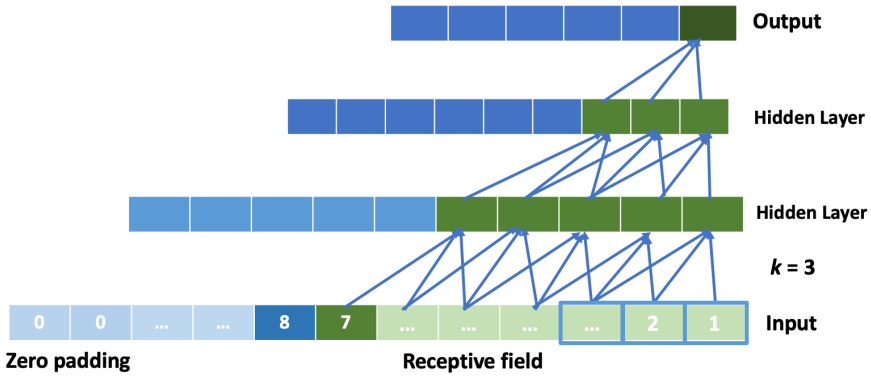
- 1D convolutions process all time steps in parallel, removing the sequential bottleneck of RNN/LSTM.
- The convolutions in the architecture are causal, meaning that there is no information leakage from future to past
- Greatly reducing vanishing/exploding gradient issues with dilation, capturing long-range dependencies more reliably
- No recurrent gates; fewer parameters to store during training and simpler architecture to tune.

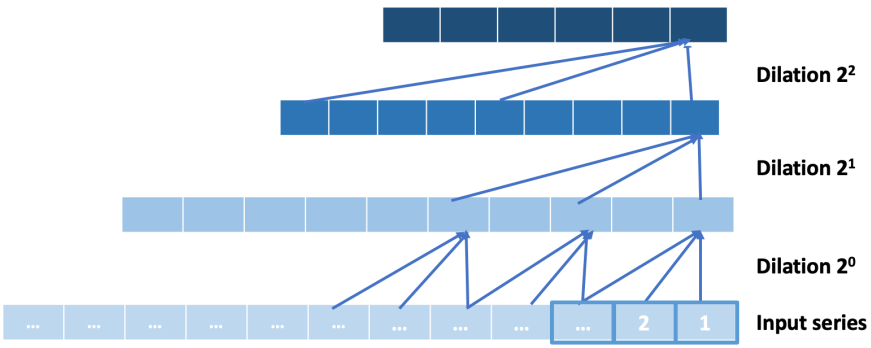
[1] Bai, Shaojie. "An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling." arXi

TCN Basic Concepts

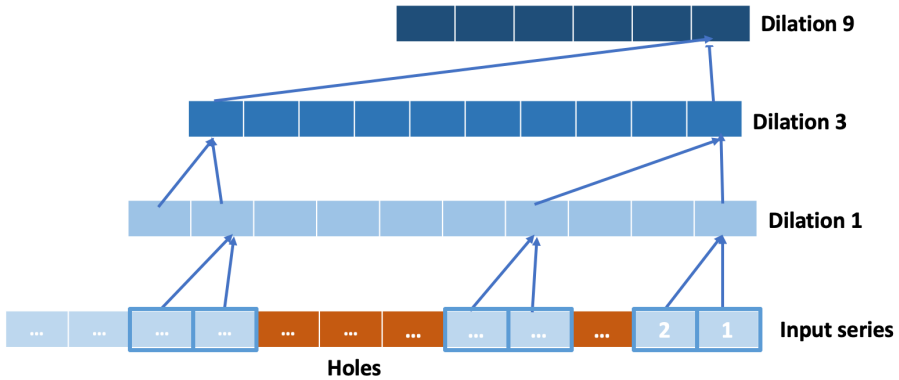
- **Temporal Convolutional Networks (TCN)** are a deep learning architecture for processing sequential 1D data by using causal, dilated 1D convolutional layers
- **Causal convolution:** every timestep in the output can get information from the current/past timesteps
- **Receptive field** RF: set of entries of the input that affect a specific entry of the output.

$$RF = 1 + l \cdot (k - 1)$$



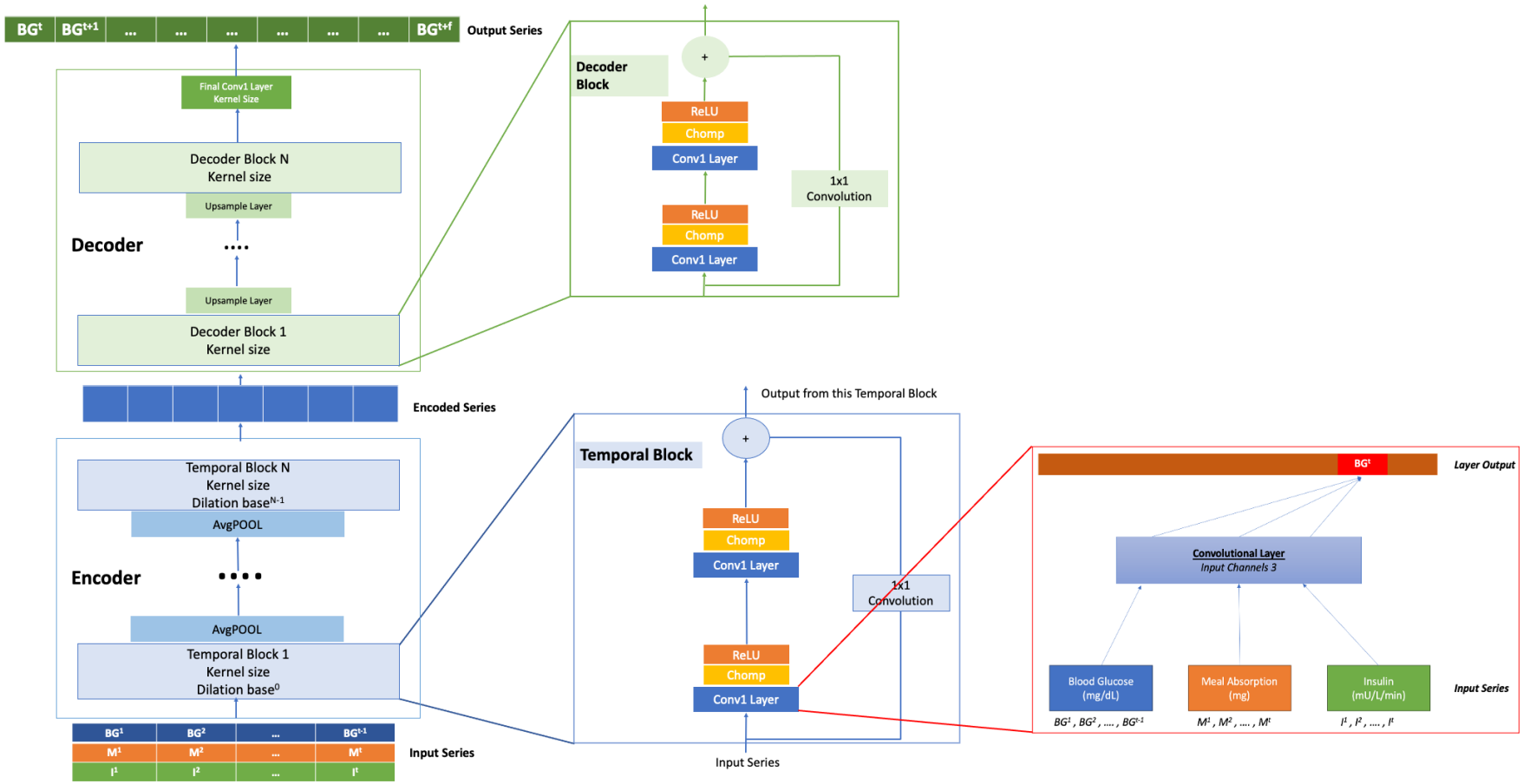


Dilation: *distance between the elements of the input sequence used to compute one entry of the output sequence.*



Hole *formulate when the dilation base b is larger than the kernel size k.* $k \leq b$

TCN for Blood Glucose Prediction



Full architecture of the proposed TCN model with temporal blocks and Convolutional layers

The **receptive field** of the TCN is given by,

$$RF = 1 + 2 \cdot \sum_{i=0}^{n-1} (k-1) \cdot b^i = 1 + 2 \cdot (k-1) \cdot \frac{b^n - 1}{b - 1} \tag{1}$$

Bayesian TCN

Let $\boldsymbol{D}$ be the training data,

$$\boldsymbol{D} = \left\{ (G^{(k-l:k)},\, M^{(k-l:k)},\, I^{(k-l:k)},\, G^{(k+1:k+K_f)}) \right\}_{k=1}^{N}.$$

Bayes' rule for the posterior:

$$\pi_{\text{post}}(\boldsymbol{w} \mid \boldsymbol{D}) = \frac{\pi_{\text{like}}(\boldsymbol{D} \mid \boldsymbol{w})\, \pi_{\text{pr}}(\boldsymbol{w})}{\pi_{\text{evid}}(\boldsymbol{D})}.$$

- Prior with σ_{pr}^2 variance,

$$\pi_{\text{pr}}(\boldsymbol{w}) = \mathcal{N}\left(0,\, \sigma_{\text{pr}}^2\right)$$

- Likelihood under Gaussian output noise:

$$\log \pi_{\text{like}}(\boldsymbol{D} \mid \boldsymbol{w}) = -\frac{1}{2\sigma_{\text{noise}}^2} \sum_{t=1}^{N} \left\| G^{(k+1:k+K_f)} - G_{\text{TCN}}^{(k+1:k+K_f)}(\boldsymbol{w}) \right\|_2^2 - \frac{NK_f}{2} \log(2\pi\sigma_{\text{noise}}^2)$$

- The predictive distribution of the future blood glucose trajectory $G^{\star(k+1:k+K_f)}$ is:

$$\pi\left(G^{\star(k+1:k+K_f)} \mid G^{\star(k-l:k)}, M^{\star(k-l:k)}, I^{\star(k-l:k)}, \boldsymbol{D}\right)$$

$$= \int \pi\left(G^{\star(k+1:k+K_f)} \mid G^{\star(k-l:k)}, M^{\star(k-l:k)}, I^{\star(k-l:k)}, \boldsymbol{w}\right) \pi_{\text{post}}(\boldsymbol{w} \mid \boldsymbol{D})\, d\boldsymbol{w} \quad (2)$$

Hierarchical Bayesian Inference

Bayesian TCN Model:

- Convolutional weight parameters: $\boldsymbol{w}$
- Inference hyperparameters : $\boldsymbol{\sigma} = [\boldsymbol{\sigma}^{\text{pr}}, \boldsymbol{\sigma}^{\text{noise}}]$
- Architecture hyperparameters : $\boldsymbol{\xi}$ (e.g. number of temporal blocks, dilation base, stride)

Hierarchical Bayesian Inference:[2]

- *Level 1.* Infer the network parameters $\boldsymbol{w}$,

$$\pi_{post}(\boldsymbol{w}|\boldsymbol{D}, \boldsymbol{\sigma}, \boldsymbol{\xi}) = \frac{\pi_{like}(\boldsymbol{D}|\boldsymbol{w}, \boldsymbol{\sigma}, \boldsymbol{\xi})\, \pi_{pr}(\boldsymbol{w}|\boldsymbol{\sigma}, \boldsymbol{\xi})}{\pi_{evid}(\boldsymbol{D}|\boldsymbol{\sigma}, \boldsymbol{\xi})}.$$

- *Level 2.* Infer the inference hyper-parameters $\boldsymbol{\sigma}$,

$$\pi_{post}(\boldsymbol{\sigma}|\boldsymbol{D}, \boldsymbol{\xi}) = \frac{\pi_{evid}(\boldsymbol{D}|\boldsymbol{\sigma}, \boldsymbol{\xi})\, \pi_{pr}(\boldsymbol{\sigma}|\boldsymbol{\xi})}{\pi_{evid}(\boldsymbol{D}|\boldsymbol{\xi})}.$$

- *Level 3.* Infer the archtecture hyper-parameters $\boldsymbol{\xi}$,

$$\pi_{post}(\boldsymbol{\xi}|\boldsymbol{D}) = \frac{\pi_{evid}(\boldsymbol{D}|\boldsymbol{\xi})\, \pi_{pr}(\boldsymbol{\xi})}{\pi_{evid}(\boldsymbol{D})}.$$

[7] Singh, Pratyush Kumar, Kathryn A. Farrell-Maupin, and Danial Faghihi. "A framework for strategic discovery of credible neural network surrogate models under uncertainty." Computer Methods in Applied Mechanics and Engineering 427 (2024): 117061.

The Evidence Framework

- Variational Inference: We approximate the posterior with a Gaussian Distribution $q_{\boldsymbol{\eta}}(\boldsymbol{w})$, where $\boldsymbol{\eta} = (\mu_\theta, \sigma_\theta)$.

$$\boldsymbol{\eta}^{\text{opt}} = \arg\min_{\boldsymbol{\eta}} \mathcal{L}_{\text{VI}}(\boldsymbol{\eta})$$

Where, (for Gaussian-distributed total error)

$$\begin{aligned}\mathcal{L}_{\text{VI}}(\boldsymbol{\eta}) &= \text{KL}\big[q_{\boldsymbol{\eta}}(\boldsymbol{w}) \,\|\, \pi_{\text{pr}}(\boldsymbol{w})\big] + \frac{Nf}{2}\log(2\pi\sigma^2_{\text{noise}}) \\ &+ \frac{1}{M_{MC}}\sum_{j=1}^{M_{MC}} \frac{1}{2\sigma^2_{\text{noise}}}\big\|G^{(k+1:k+K_f)} - G^{(k+1:k+K_f)}_{\text{TCN}}(\hat{\boldsymbol{w}}^{(j)})\big\|_2^2 \end{aligned} \tag{3}$$

- The optimal values $\boldsymbol{\sigma}^{\text{opt}}$ correspond to the Maximum A Posteriori (MAP) estimate:

$$\boldsymbol{\sigma}^{\text{opt}} = \operatorname*{argmax}_{\boldsymbol{\sigma}} \mathcal{L}_{\text{Evid2}}(\boldsymbol{\sigma}) \tag{4}$$

The log-evidence for fixed $\boldsymbol{\xi}$ can be analytically evaluated as

$$\begin{aligned}\mathcal{L}_{\text{Evid2}}(\boldsymbol{\sigma}) = \log \pi_{\text{evid}}(\boldsymbol{D} \mid \boldsymbol{\sigma}, \boldsymbol{\xi}) \approx &-\frac{1}{2\sigma^2_{\text{noise}}}\sum_{t=1}^{N}\big\|G^{(k+1:k+K_f)} - G^{(k+1:k+K_f)}_{\text{TCN}}(\boldsymbol{\mu}_{\text{post}})\big\|_2^2 \\ &- \frac{Nf}{2}\log(2\pi\sigma^2_{\text{noise}}) - \frac{1}{2\sigma^2_{\text{pr}}}\|\boldsymbol{\mu}_{\text{post}}\|_2^2 \\ &- \frac{p}{2}\log(2\pi\sigma^2_{\text{pr}}) - \frac{1}{2}\log\det(\boldsymbol{H})\end{aligned} \tag{5}$$

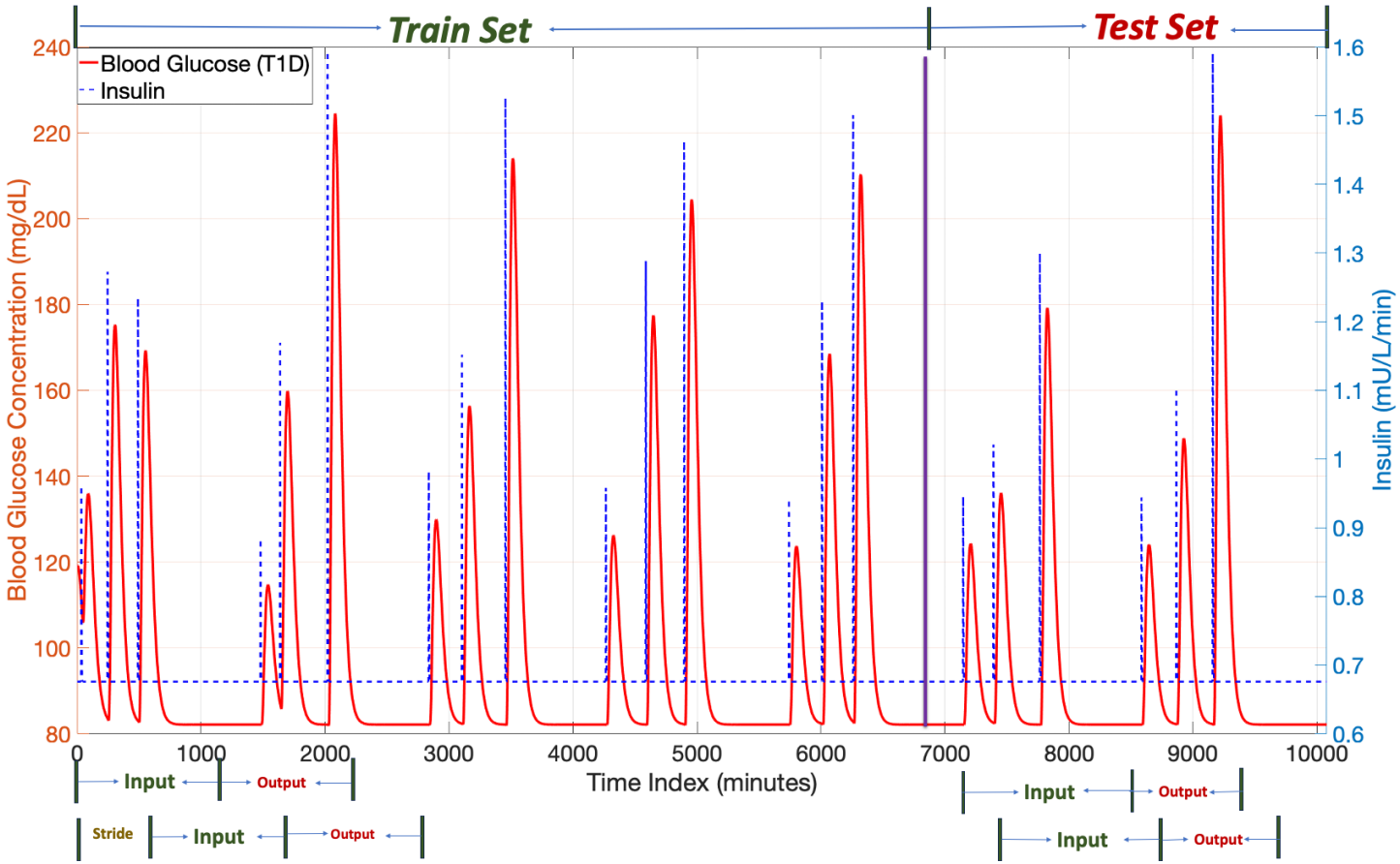
- The hyperparameters $\boldsymbol{\sigma}$ around their MAP values $\boldsymbol{\sigma}^{\text{opt}}$,

$$\text{Evid}_3 = \pi_{\text{evid}}(\boldsymbol{D}|\boldsymbol{\xi}) = \pi_{\text{evid}}(\boldsymbol{D}|\boldsymbol{\sigma}^{\text{opt}}, \boldsymbol{\xi})\, \pi_{\text{pr}}(\boldsymbol{\sigma}^{\text{opt}}|\boldsymbol{\xi}) \left[2\pi\sqrt{\text{Var}(\sigma^2_{\text{pr}})\text{Var}(\sigma^2_{\text{noise}})}\right] \tag{6}$$

TCN Training

Training Setup:

- Induced Uncertainty in meal time about ± 60 minitues and in meal size $\pm 60\%$.
- 5 days training data and 2 days of testing/validation data.
- Input length 1 day and output length is 12 hours.
- Data was generated with a stride of 3.



Synthetic Training Data

Bergman Minimal Model[3]

$$\dot{G}(t) = -(X(t) + p_1)G(t) + p_1 G_b + \frac{R_{ag}(t)}{V_g}$$

$$\dot{X}(t) = -p_2 X(t) + p_3(I(t) - I_b)$$

$$\dot{I}(t) = \begin{cases} -p_4 I(t) + \gamma(G(t) - h)(t - t_m), & G(t) \geq h,\ t \geq t_m \\ -p_4 I(t), & \text{otherwise} \end{cases}$$

Gut Absorption Model (Dalla Man)

$$R_{ag}(t) = f\, k_{\text{abs}}\, q_{\text{gut}}(t)$$

$$\dot{q}_{\text{gut}} = -k_{\text{abs}} q_{\text{gut}} + k_{\text{empt}} q_{\text{sto2}}$$

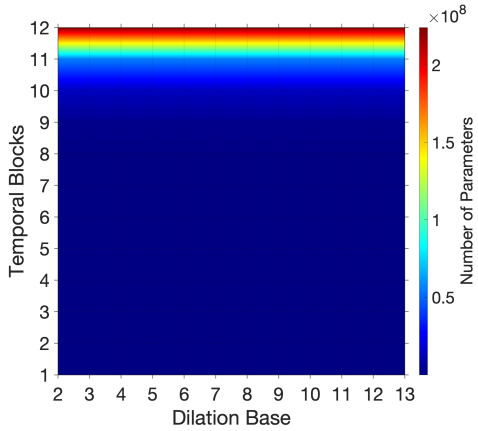
$$\dot{q}_{\text{sto1}} = -k_{21} q_{\text{sto1}} + M\delta(t - t_m), \qquad \dot{q}_{\text{sto2}} = -k_{\text{empt}} q_{\text{sto2}} + k_{21} q_{\text{sto1}}$$

Key states and parameters:

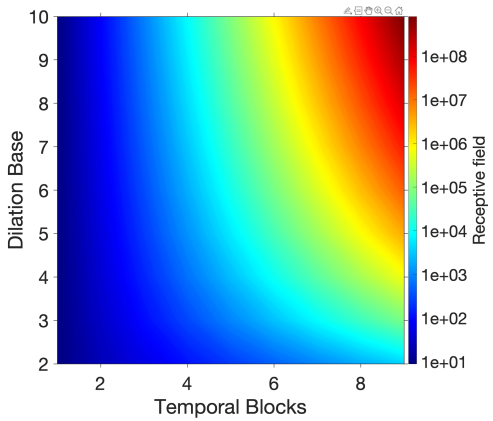
- $G(t)$: plasma glucose, $X(t)$: insulin action, $I(t)$: plasma insulin
- p_1: glucose effectiveness, p_2, p_3: insulin action dynamics, p_4: insulin decay
- G_b, I_b: basal glucose/insulin, h: secretion threshold, γ: secretion gain
- $R_{ag}(t)$: gut-driven glucose appearance, V_g: glucose distribution volume

[3] Nandi, Souransu, and Tarunraj Singh. "Differential flatness based run-to-run control of blood glucose for people with type 1 diabetes."

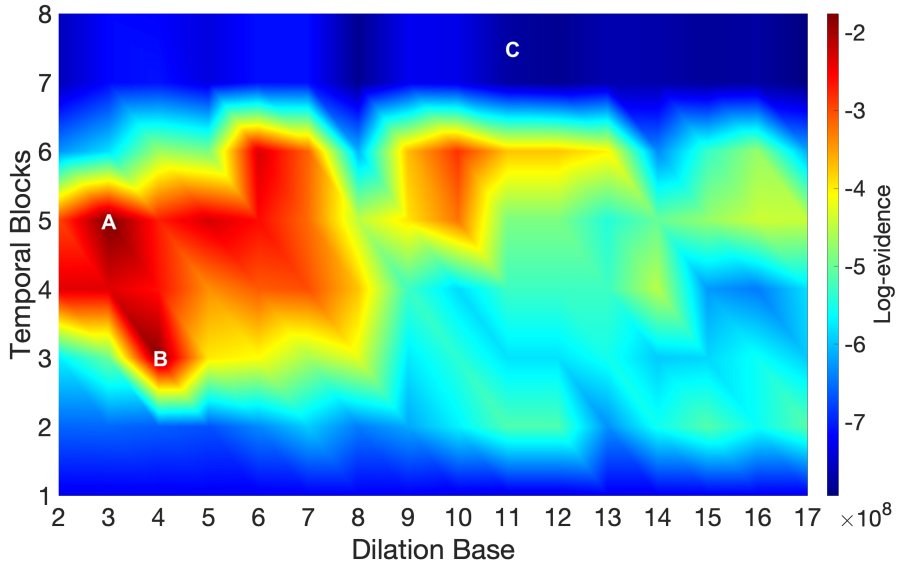
Numerical Results I



Number of parameters

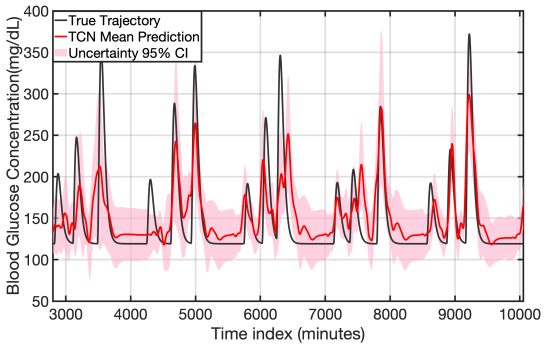


Receptive field

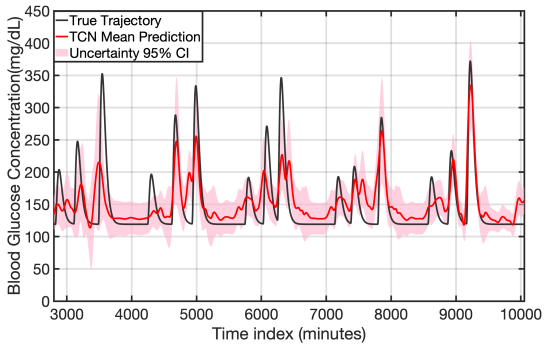


Model evidence (posterior model plausibility)

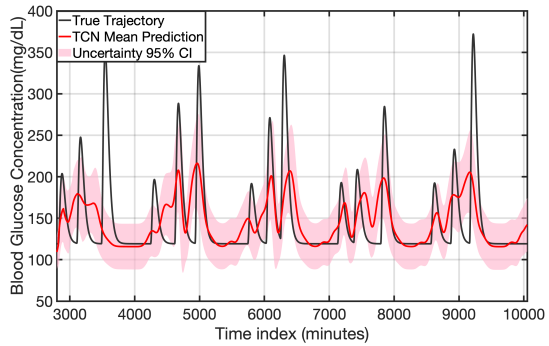
Numerical Results II



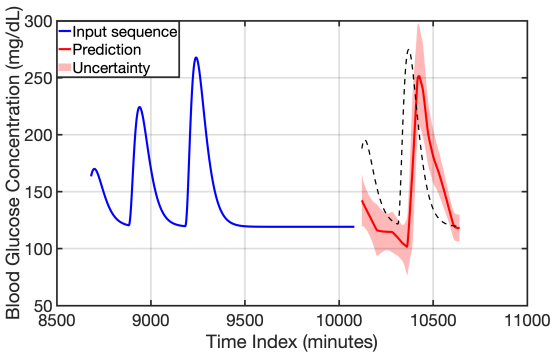
Training data fit for A



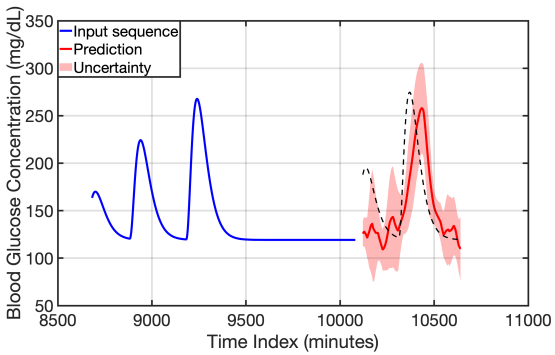
Training data fit for B



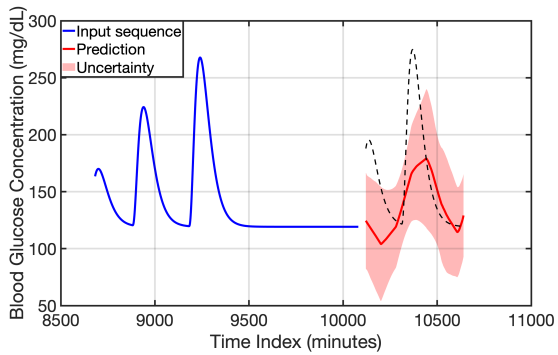
Training data fit for C



Prediction for A

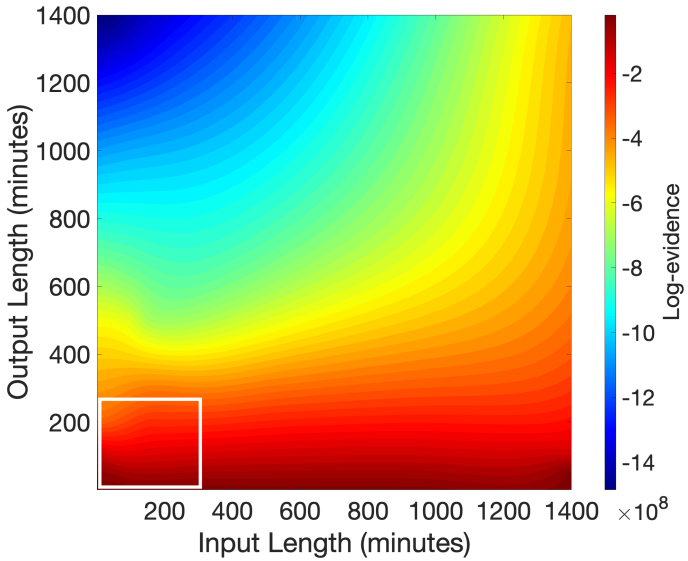


Prediction for B

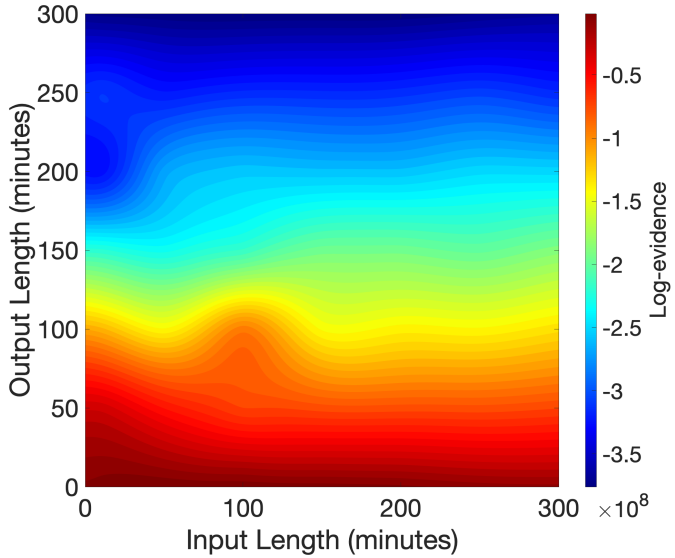


Prediction for C

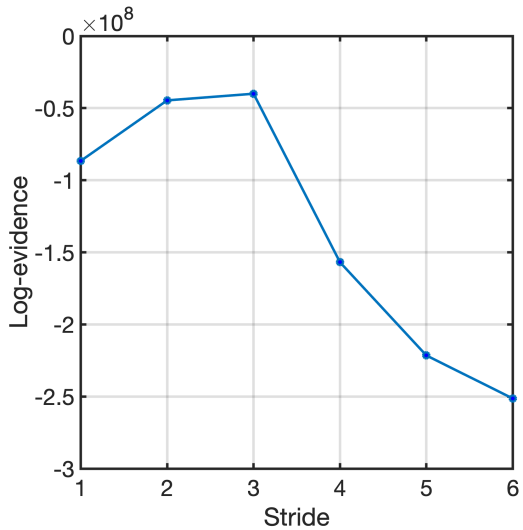
Numerical Results III



Resolution 1400×1400



Resolution 300×300



Stride

Outline

1 Motivation

2 Bayesian Temporal Convolutional Network (B-TCN)

3 B-TCN Model Discovery Using Population Based Data

4 Future Plan with Digital Twin for Diabetes

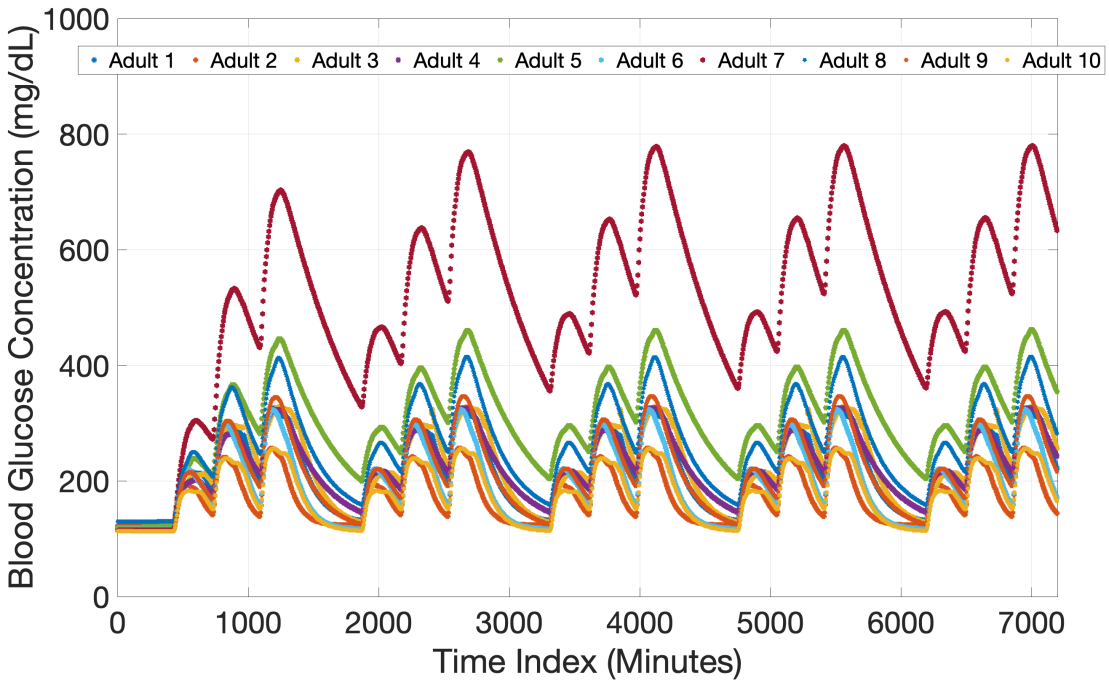
Population Based Patient Data

Training Setup:

- Input length: 1 day, Output length: 285 minutes (minimum inter-meal gap).

Validation Setup:

- 3 days for training and 1 day for validation.
- Input length: 1 day and output length: 60 minutes.



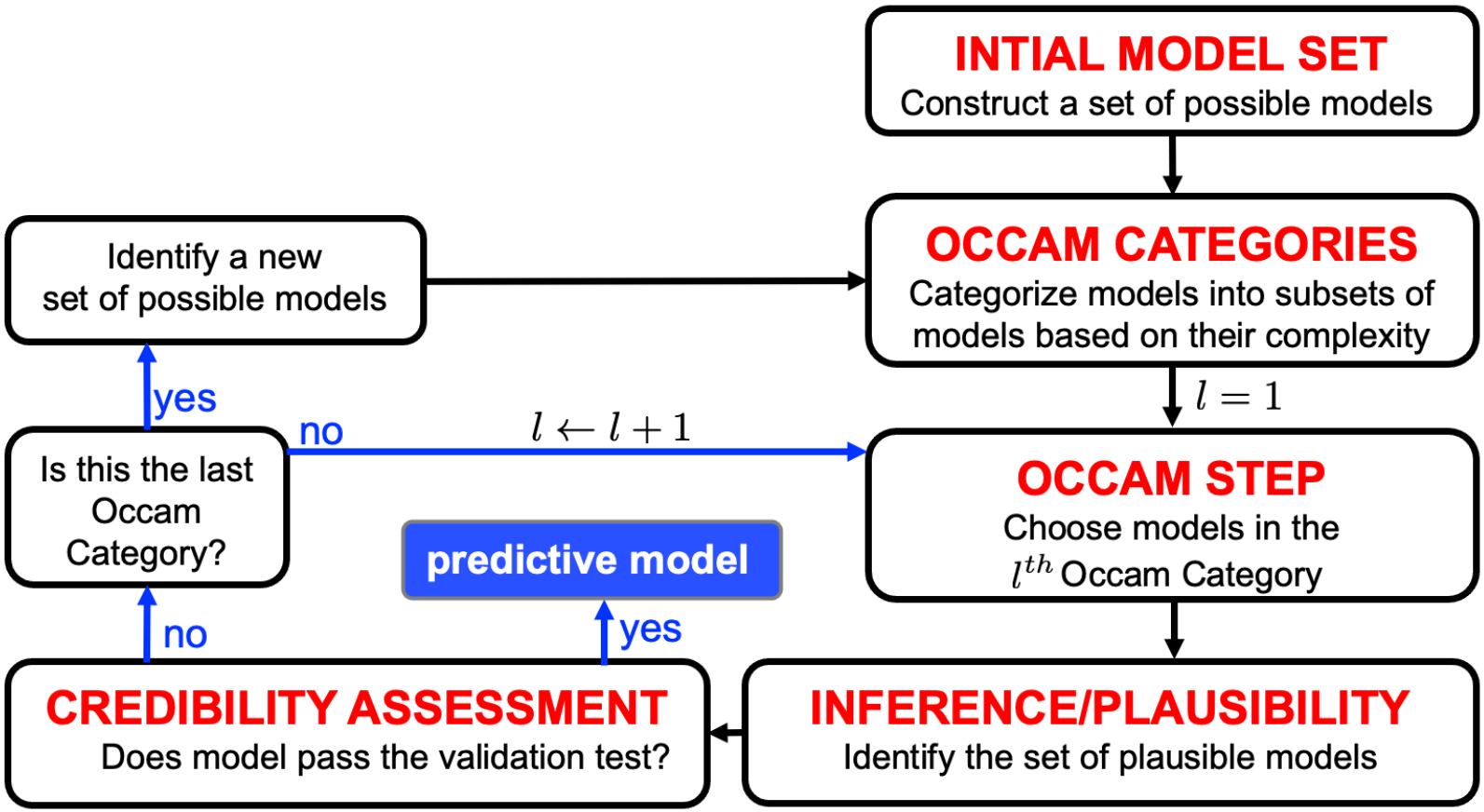
Population based patient data from UVa/Padova T1D simulator

5-fold cross-validation:

- Fold-1: Training patinets: Adults [1-8], validation patients: Adults [9,10].
- Fold-2: Training patinets: Adults [1-5,8,9,10], validation patients: Adults [6,7].
- Fold-3: Training patinets: Adults [3-10], validation patients: Adults [1,2].
- Fold-4: Training patinets: Adults [1,2,3-10], validation patients: Adults [3,4].
- Fold-5: Training patinets: Adults [1-4,6,7,9,10], validation patients: Adults [5,8].

Bayesian Model Selection Algorithm (BMSA)

Discovering B-TCN architecture using population-based data of multiple patients



Identifying Initial Pool of TCN Architectures

1. The receptive field has to be greater than or equal to one day,

$$1440 \leq RF_n \leq 2000 \tag{7}$$

2. Condition for no holes: $k \geq b$.

Additional assumptions:

- Number of filters in each block starts from 2 and can go up as much as 512.
- Upper bound on kernel size 15.
- We double the number of filters in each temporal block[4] [5].

Category	Params	ArchCount	# blocks n	# filters f	# (k, b)
1	$\leq$30k	10	[3, 4]	[2, 4]	6
2	30k-100k	9	[3, 4, 5, 6]	[2, 4, 8]	9
3	100k-400k	10	[3, 4, 5, 6]	[2, 4, 8, 16]	10
4	400k-1M	11	[3, 4, 5, 6, 7]	[2, 4, 8, 16, 32]	11
5	1M-5M	11	[3, 4, 5, 6, 7]	[4, 8, 16, 32, 64]	11
6	$\geq$5M	10	[3, 4, 5, 6, 7]	[8, 16, 32, 64, 128]	10

[4] Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. "U-net: Convolutional networks for biomedical image segmentation." International Conference on Medical image computing and computer-assisted intervention. Cham: Springer international publishing, 2015.

[5] Badrinarayanan, Vijay, Alex Kendall, and Roberto Cipolla. "Segnet: A deep convolutional encoder-decoder architecture for image segmentation." IEEE transactions on pattern analysis and machine intelligence 39.12 (2017): 2481-2495.

Bayesian TCN Validation Metrics

B-TCN: 1 hour prediction of blood glucose

- B-TCN mean prediction $\hat{G}^{\text{TCN}}$ and prediction uncertainty (variance): σ^{TCN}
- Blood glucose measurements: G^{data}
- Sensor noise σ^{data}: manufacturing-specific of CGM (e.g., Dexcom), FDA-approved code provides true vs measured blood glucose

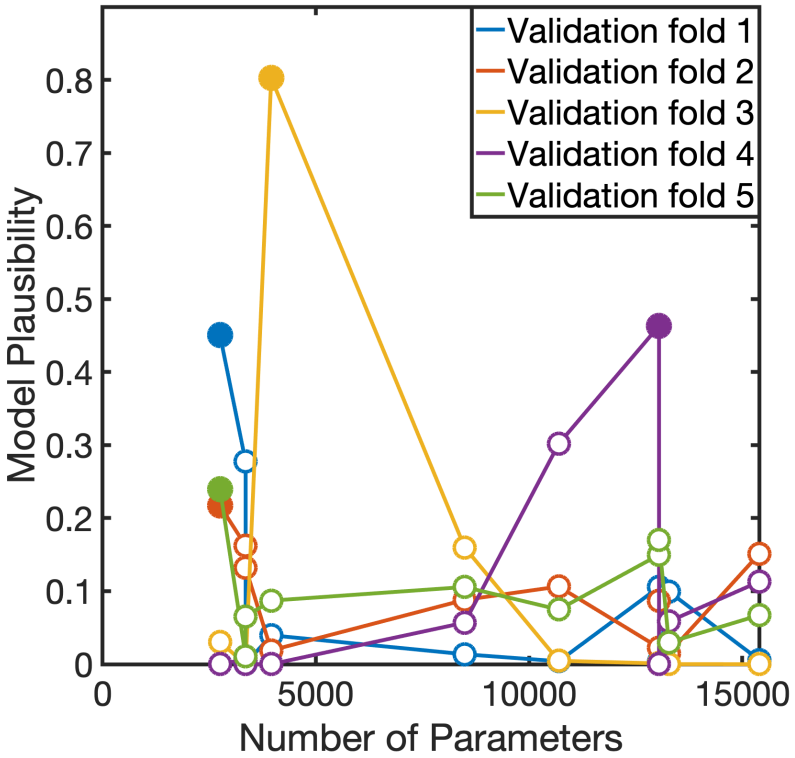
Negative log-prediction density

$$\sigma^2 = (\sigma^{\text{TCN}})^2 + (\sigma^{\text{data}})^2$$

$$NLPD = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{\left(\hat{G}_i^{\text{TCN}} - \hat{G}_i^{\text{data}}\right)^2}{\sigma_i^2} + \ln(\sigma_i^2)\right] \leq Tol$$

$\hat{G}_i^{\text{TCN}}$ represents the Bayesian TCN mean prediction of blood glucose at time i, while $\hat{G}_i^{\text{data}}$ denotes the corresponding CGM-measured glucose value. The predictive uncertainty consists of the model-derived standard deviation σ_i^{TCN} and the CGM sensor noise σ^{data}.

Category 1: Most Plausible Models



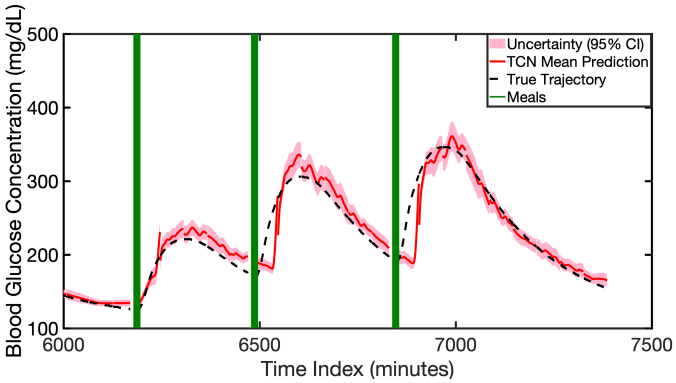
Model plausibility of architectures in Category 1 for 3 different cross-validation folds

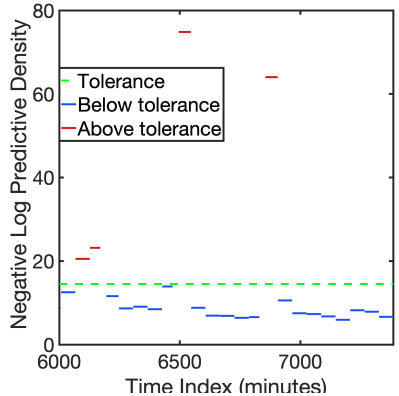
Most plausible architectures (Category 1):

- **Fold 1:** Dilation Base = 9, Encoder = [2,4,8], Decoder = [8,4,2], kernel size: 9
- **Fold 2:** Dilation Base = 9, Encoder = [2,4,8], Decoder = [8,4,2], kernel size: 9
- **Fold 3:** Dilation Base = 8, Encoder = [2,4,8], Decoder = [8,4,2], kernel size: 13
- **Fold 4:** Dilation Base = 9, Encoder = [4,8,16], Decoder = [16,8,4], kernel size: 9
- **Fold 5:** Dilation Base = 9, Encoder = [2,4,8], Decoder = [8,4,2], kernel size: 11

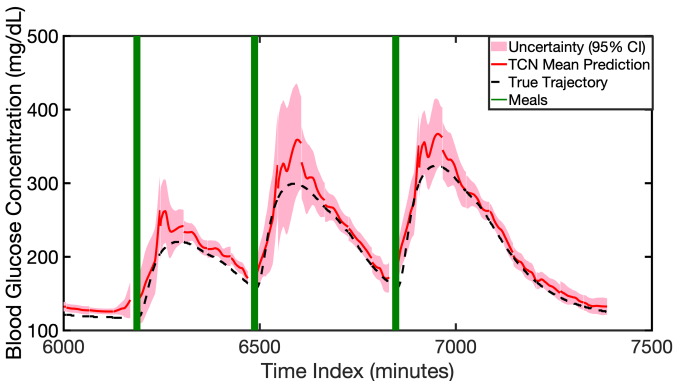
Category 1: Validation Test

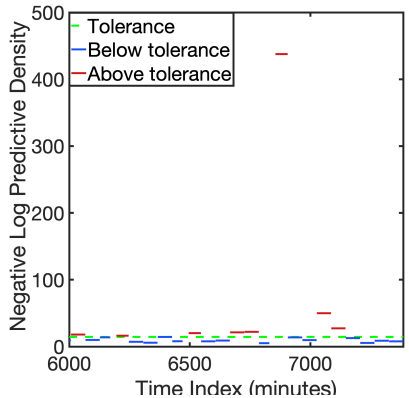
Fold 1
Adult 9



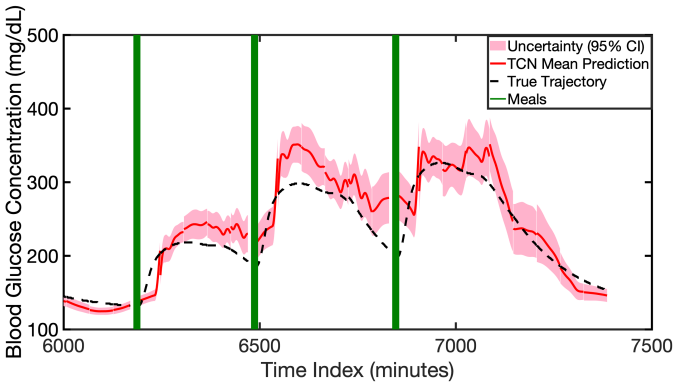


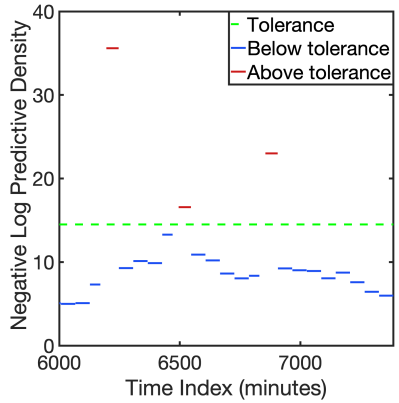
Fold 2
Adult 6



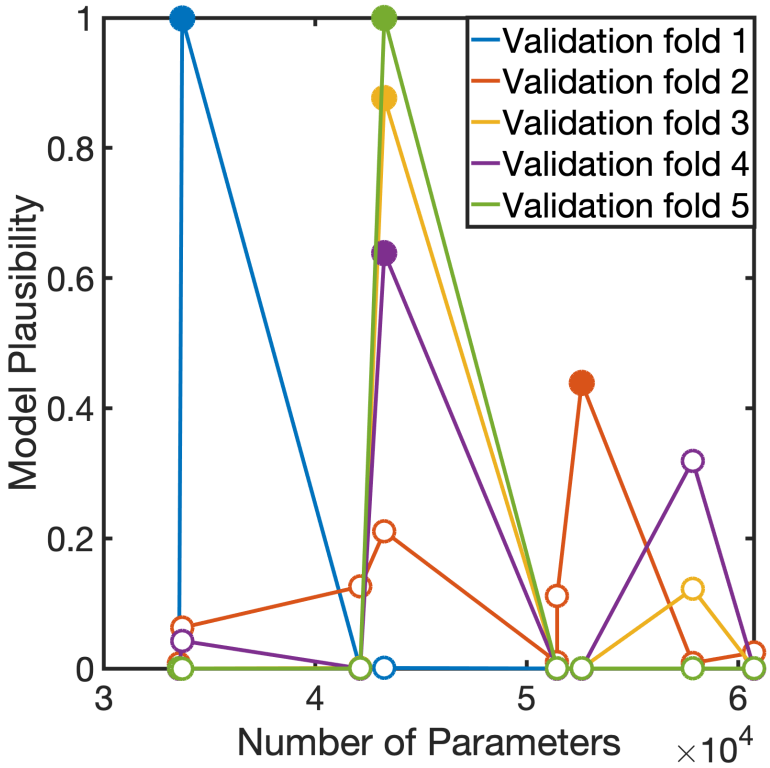


Fold 3
Adult 1





Category 2: Most Plausible Models



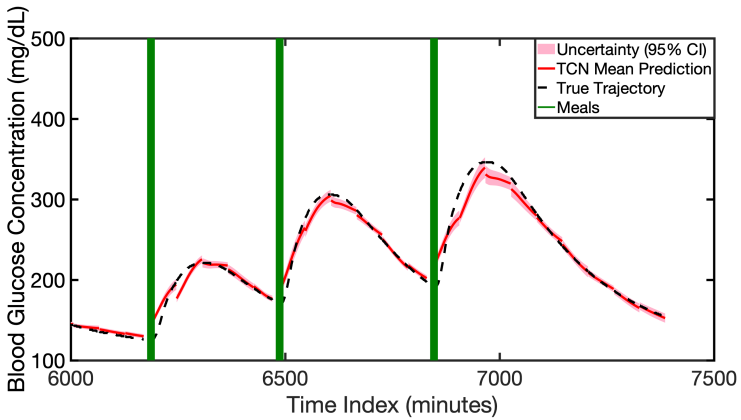
Model plausibility calculated against number of parameters for 3 different validation sets for Category 2

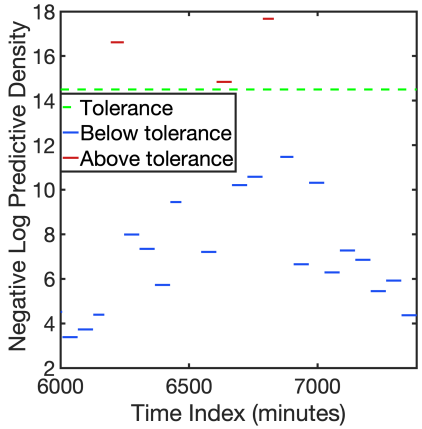
Most plausible architectures (Category 2):

- **Fold 1:**Dilation Base = 3 Encoder = [2,4,8,16,32] Decoder = [32,16,8,4,2] kernel size: 7
- **Fold 2:**Dilation Base = 4 Encoder = [4,8,16,32] Decoder = [32,16,8,4] kernel size: 11
- **Fold 3:**Dilation Base = 3 Encoder = [2,4,8,16,32] Decoder = [32,16,8,4,2] kernel size: 9
- **Fold 4:**Dilation Base = 3 Encoder = [2,4,8,16,32] Decoder = [32,16,8,4,2] kernel size: 9
- **Fold 5:**Dilation Base = 3 Encoder = [2,4,8,16,32] Decoder = [32,16,8,4,2] kernel size: 9

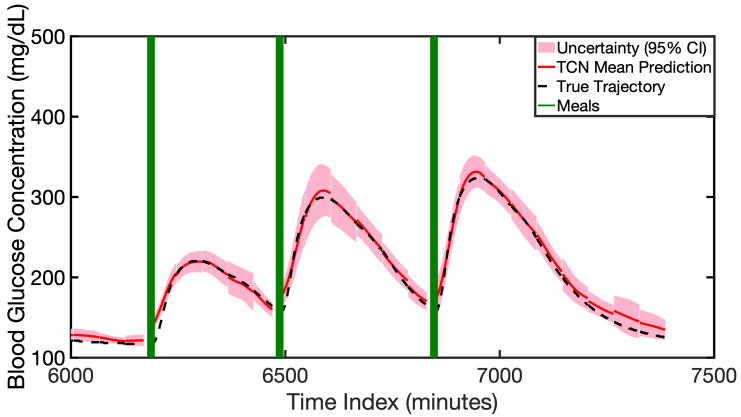
Category 2: Validation Test (Fail)

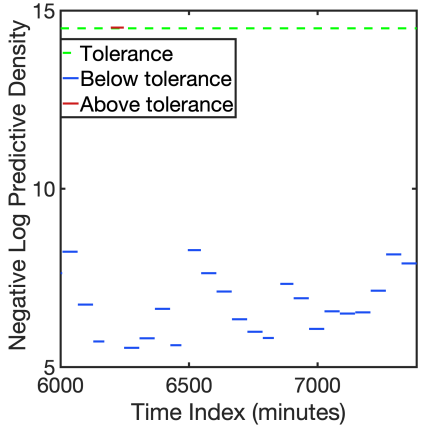
Fold 1
Adult 9





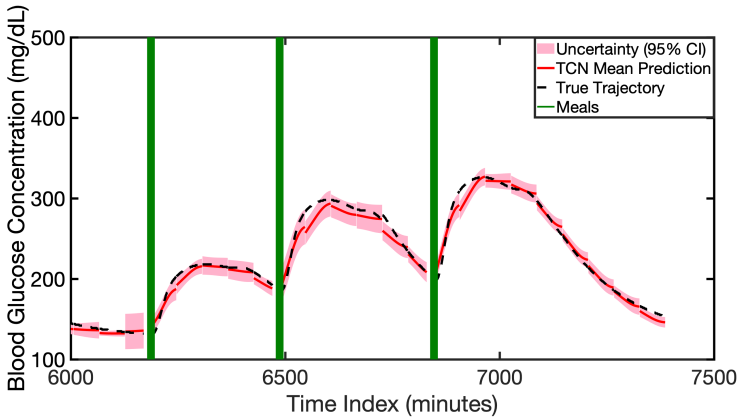
Fold 2
Adult 6

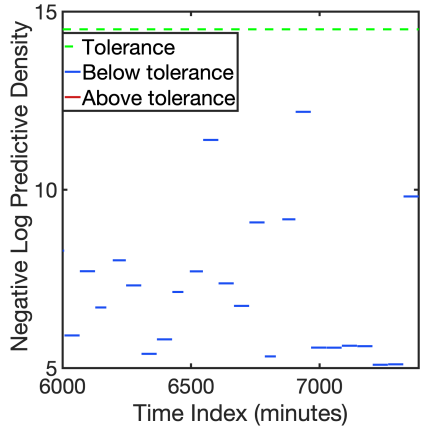




Category 2: Validation Test (Success)

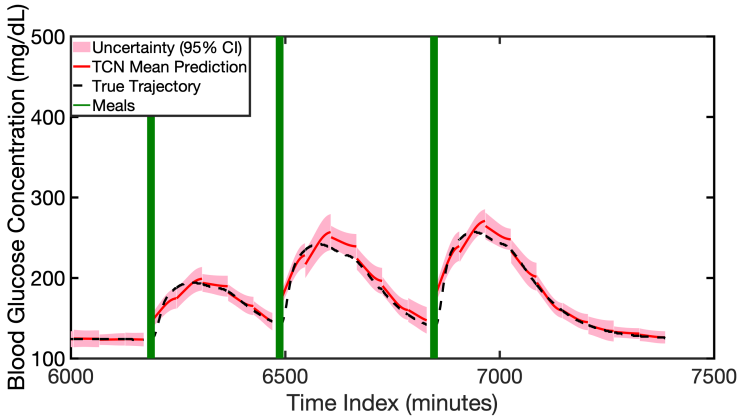
Fold 3
Adult 1

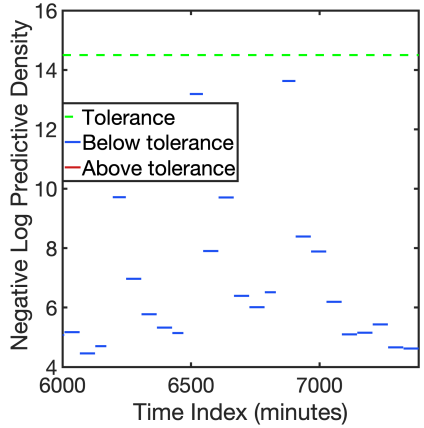




Valid Architecture: Dilation Base = 3 Encoder = [2,4,8,16,32] Decoder = [32,16,8,4,2] kernel size: 9

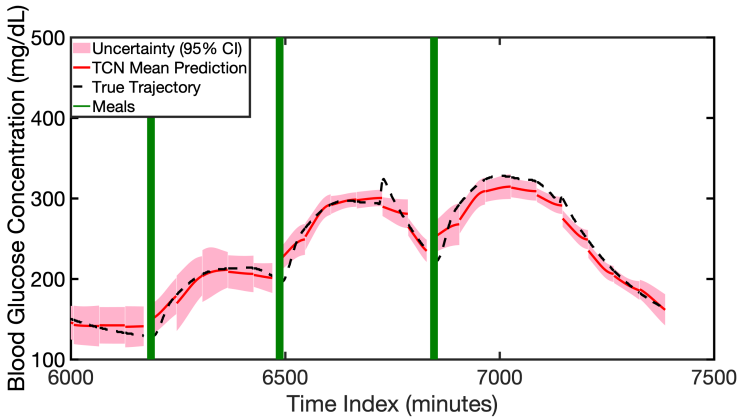
Fold 3
Adult 2

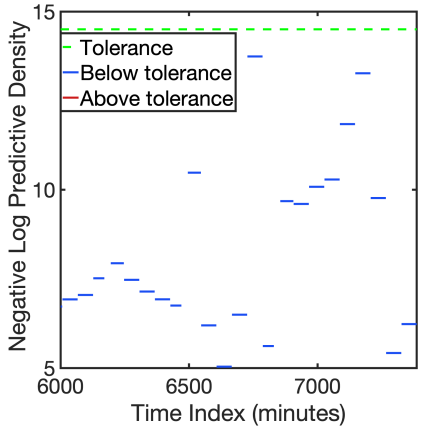




Category 2: Validation Test (Success)

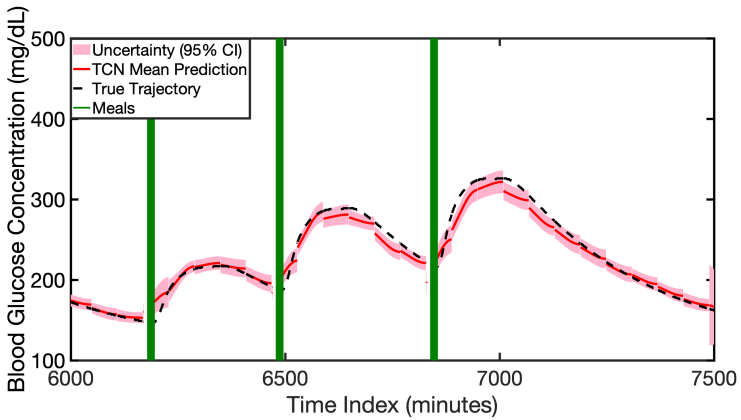
Fold 4
Adult 3

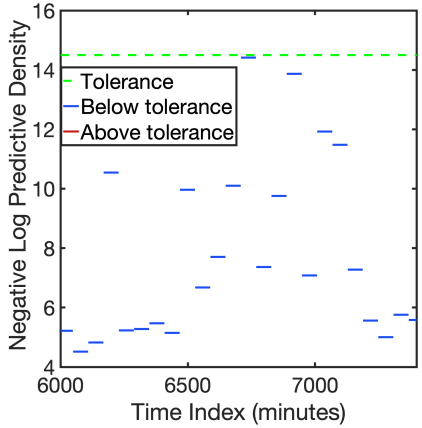




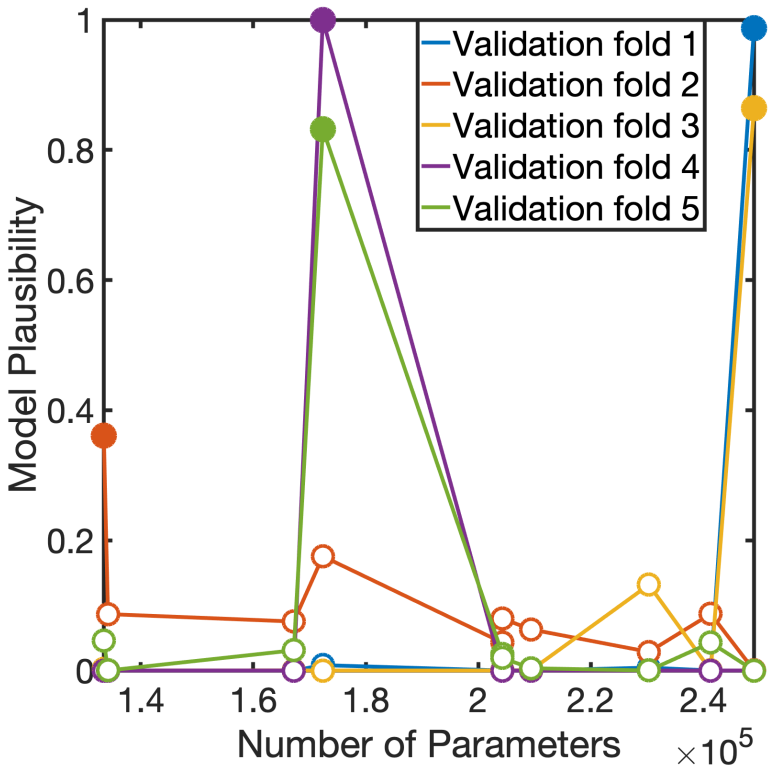
Valid Architecture: Dilation Base = 3 Encoder = [2,4,8,16,32] Decoder = [32,16,8,4,2] kernel size: 9

Fold 4
Adult 4





Category 3: Most Plausible Models



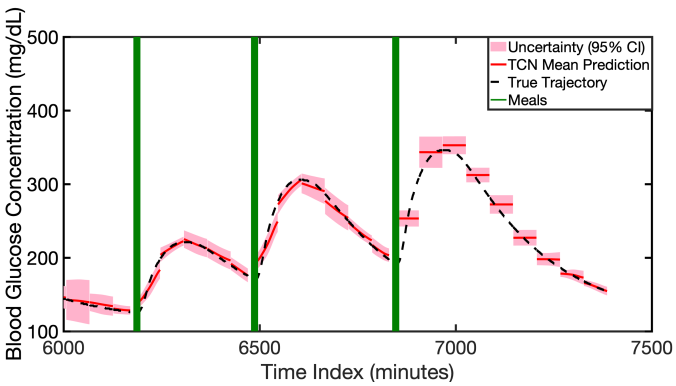
Model plausibility calculated against number of parameters for 3 different validation sets for Category 3

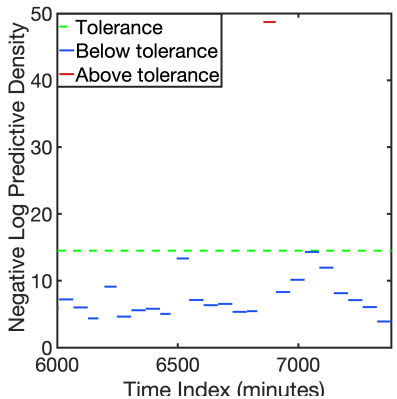
Most plausible architectures (Category 3):

- **Fold 1:**Dilation Base = 2 Encoder = [2,4,8,16,32,64] Decoder = [64,32,16,8,4,2] kernel size: 13
- **Fold 2:**Dilation Base = 9 Encoder = [16,32,64] Decoder = [64,32,16] kernel size: 9
- **Fold 3:**Dilation Base = 2 Encoder = [2,4,8,16,32,64] Decoder = [64,32,16,8,4,2] kernel size: 13
- **Fold 4:**Dilation Base = 3 Encoder = [4,8,16,32,64] Decoder = [64,32,16,8,4] kernel size: 9
- **Fold 4:**Dilation Base = 3 Encoder = [4,8,16,32,64] Decoder = [64,32,16,8,4] kernel size: 9

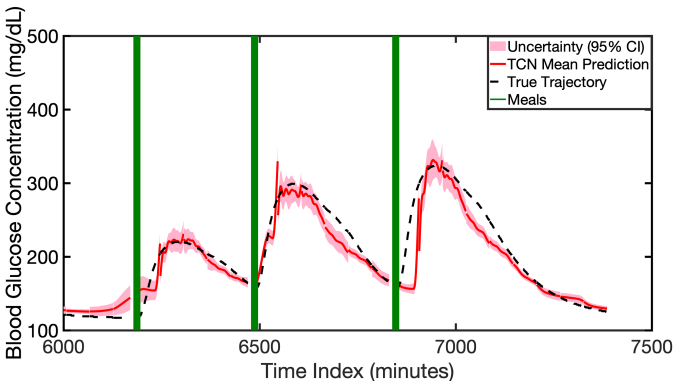
Category 3: Validation Test

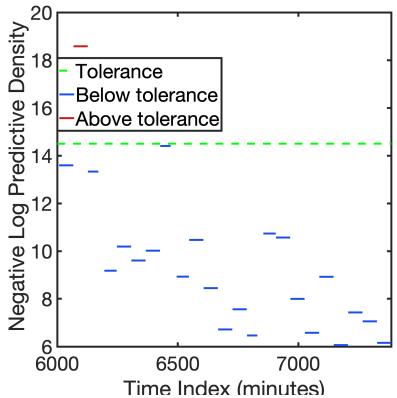
Fold 1
Adult 9



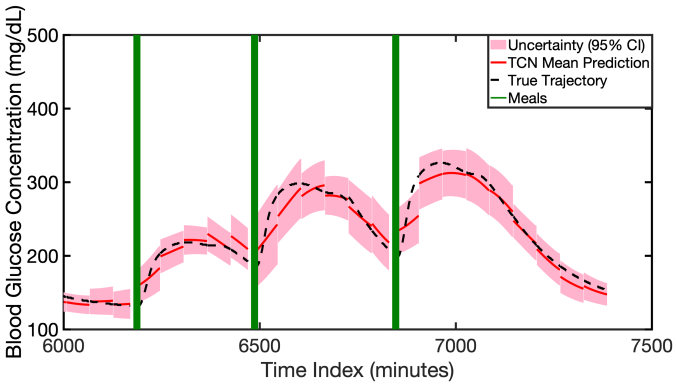


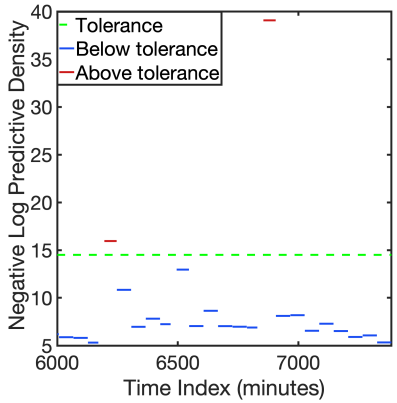
Fold 2
Adult 6





Fold 3
Adult 1





Outline

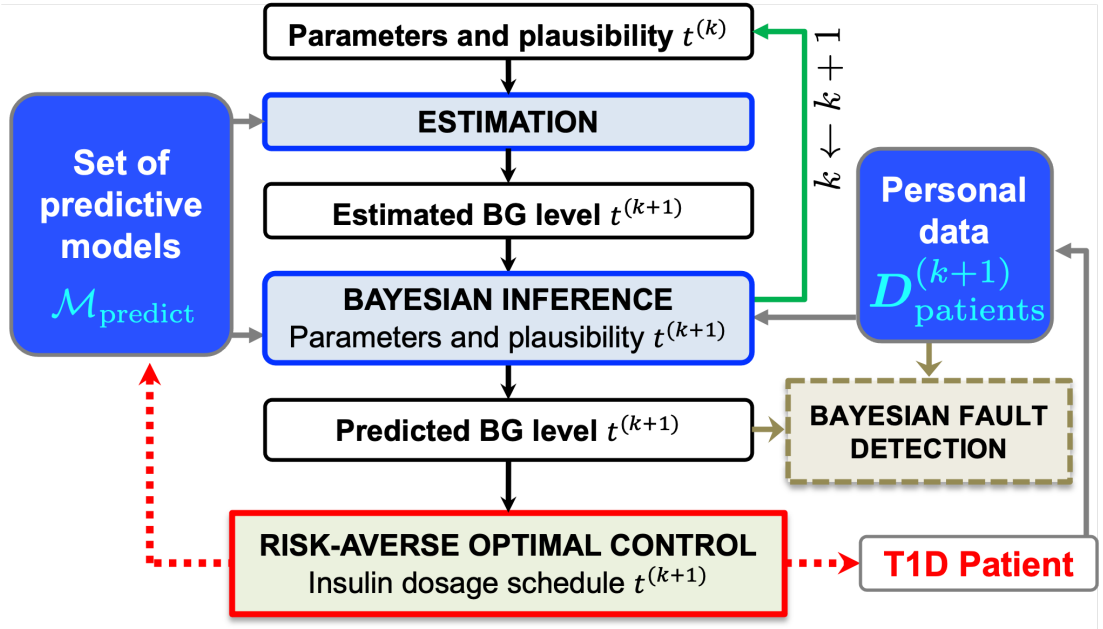
1 Motivation

2 Bayesian Temporal Convolutional Network (B-TCN)

3 B-TCN Model Discovery Using Population Based Data

4 Future Plan with Digital Twin for Diabetes

Digital Twin (DT)



The DT learns blood-glucose (BG) dynamics from individualized continuous glucose monitoring (CGM) time series data and computes insulin dosing schedules that are optimized under uncertainty. The computational stack comprises below modules:

1. **Bayesian Filter:** dynamically update both weights $\boldsymbol{w}$ and inference hyperparameters $\boldsymbol{\sigma}$ as new data arrives sequentially.
2. **Risk-averse optimal control:** computation of insulin dosing/timing that minimize deviation of the predicted BG trajectory from a reference for type-1 diabetes (T1D), with mean-variance objective function.

Bayesian Filter

In a patient-specific digital twin, sequential data $\boldsymbol{D}^{(1)}, \boldsymbol{D}^{(2)}, \ldots$ arrive at times $t^{(1)}, t^{(2)}, \ldots$, with insulin input $I^{(k)}$ influencing blood glucose (BG) evolution.
A latent state can be defined,

$$\boldsymbol{v} = [\boldsymbol{w}, \boldsymbol{\sigma}]$$

$\boldsymbol{v}$ latent state, including TCN weights $\boldsymbol{w}$ and inference hyperparameters $\boldsymbol{\sigma}$.

Bayesian filter: At step $k+1$, the posterior is:

$$\pi_{\text{post}}(\boldsymbol{v}^{(k+1)} \mid \boldsymbol{D}^{(1:k+1)}) = \frac{\pi_{\text{like}}(\boldsymbol{D}^{(k+1)} \mid \boldsymbol{v}^{(k+1)}) \cdot \pi_{\text{prior}}(\boldsymbol{v}^{(k+1)} \mid \boldsymbol{D}^{(1:k)})}{\pi_{\text{evid}}(\boldsymbol{D}^{(k+1)} | \boldsymbol{D}^{(1:k)})}$$

Recursive update:

$$\pi_{\text{post}}(\boldsymbol{v}^{(k)} \mid \boldsymbol{D}^{(1:k)}) \longrightarrow \pi_{\text{post}}(\boldsymbol{v}^{(k+1)} \mid \boldsymbol{D}^{(1:k+1)})$$

Assumptions:

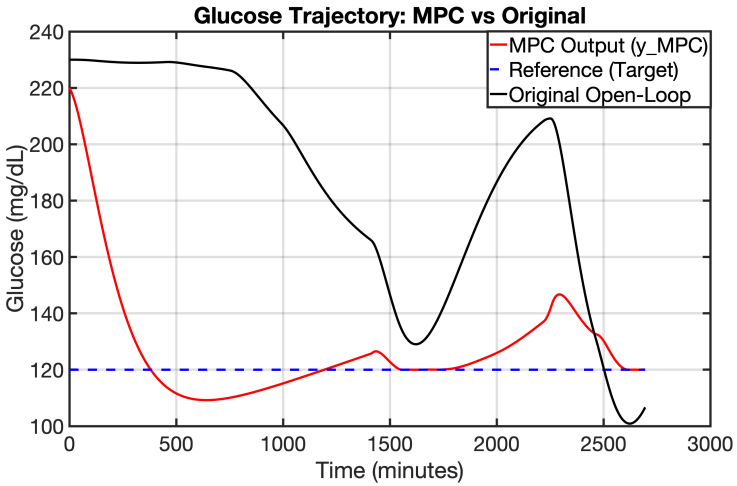
The latent state follows a first-order Markov process,

$$\pi(\boldsymbol{v}^{(k+1)} \mid \boldsymbol{v}^{(1:k)}) = \pi(\boldsymbol{v}^{(k+1)} \mid \boldsymbol{v}^{(k)}, I^{(k)}),$$

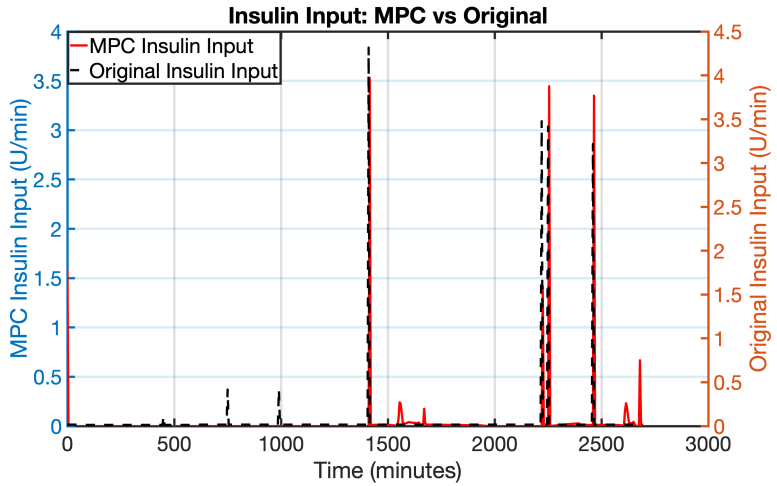
Current measurements are conditionally independent of past measurements given the current state,

$$\pi_{\text{like}}(\boldsymbol{D}^{(k+1)} \mid \boldsymbol{D}^{(1:k)}, \boldsymbol{v}^{(k+1)}) = \pi_{\text{like}}(\boldsymbol{D}^{(k+1)} \mid \boldsymbol{v}^{(k+1)}),$$

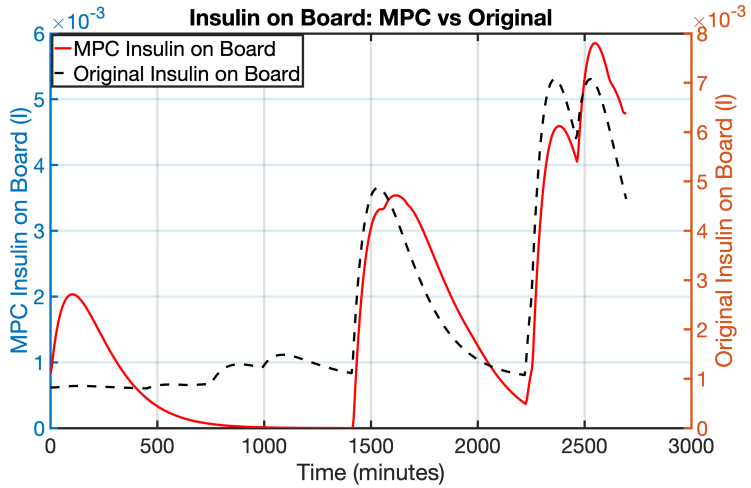
Deterministic Model Predictive Control (MPC)



MPC controlled glucose trajectory vs reference trajectory vs open loop trajectory



MPC controlled insulin input vs original insulin input



Insulin level over time for MPC controlled and original

Optimal Control Under Uncertainty

The Bayesian TCN induces a predictive distribution of future glucose

$$\boldsymbol{G}^{(k+1:k+K_f)} \sim p\Big(\boldsymbol{G}^{(k+1:k+K_f)} \mid \boldsymbol{D}^{(1:k)}, \boldsymbol{I}^{(k)}, \boldsymbol{v}^{(k)}\Big),$$

from which we compute (e.g. by Monte Carlo over $\boldsymbol{v}$) for each step $j = 1, \ldots, K_f$

$$\mu_{G,k+j}(\boldsymbol{I}^{(k)}) = \mathbb{E}\Big[G^{(k+j)}\Big], \qquad \sigma^2_{G,k+j}(\boldsymbol{I}^{(k)}) = \mathrm{Var}\Big[G^{(k+j)}\Big].$$

Chance-constrained insulin optimization problem at time k.
Let G_{lb} and G_{ub} be hypo-/hyper-glycemic thresholds, and G^{targ}_{k+j} a target trajectory (e.g. euglycemic).
We choose $\boldsymbol{I}^{(k)}$ by solving

$$\begin{aligned}
\boldsymbol{I}^{(t+1:t+f)}_{\mathrm{opt}} = \underset{\boldsymbol{I}}{\mathrm{argmin}} \quad & \Big\| \mathbb{E}\Big[G^{t+1:t+f} - G^{t+1:t+f}_{\mathrm{target}}\Big] + \beta\, \mathbb{V}\Big[G^{t+1:t+f} - G^{t+1:t+f}_{\mathrm{target}}\Big] \Big\|_2 \\
\text{subject to} \quad & \mathcal{P}\Big\{ G^{t+1:t+f} \geq G_{lb} \Big\} \geq 1 - \epsilon_1, \\
& \mathcal{P}\Big\{ G^{t+1:t+f} \leq G_{ub} \Big\} \geq 1 - \epsilon_2, \quad \text{for } (t^i_M + 120\,\mathrm{min}) \leq t \leq t^{i+1}_M, \\
& I_{BA}(t) + I_{BO}(t) \leq I_{\max}.
\end{aligned} \tag{8}$$

Here, $\boldsymbol{I} = [I_{BA}, I_{BO}]$ are the basal and bolus insulin inputs, $G^{t+1:t+f}$ is the predicted BG trajectory over the horizon f, and $G^{t+1:t+f}_{\mathrm{target}}$ is the nominal BG profile of a healthy individual. The operators $\mathbb{E}[\cdot]$ and $\mathbb{V}[\cdot]$ denote expectation and variance, respectively, with respect to the B-TCN predictive distribution.

Publications

List of current publications

1. Islam, M. A., Deighan, D., Bhattacharjee, S., Tantalo, D., Singh, P. K., Salac, D., Faghihi, D. (2025). Stochastic deep learning surrogate models for uncertainty propagation in microstructureâproperties of ceramic aerogels. Computational Materials Science, 258, 114035.
2. Islam, Md Azharul, Dwyer Scout Deighan, and Danial Faghihi. "Predicting Microstructure-Property of Silica Aerogel Materials via Bayesian Convolutional Neural Networks Surrogate Model." ASME International Mechanical Engineering Congress and Exposition. Vol. 88681. American Society of Mechanical Engineers, 2024.

Plan for future publications

1. Predictive model discovery for Digital Twin using Occum Plausibility Algorithm.
2. Digital Twin implementation with dynamic model selection.
3. Optimal insulin control for Digital twin under uncertainty.

Thank You!